# EXCEL BASIC

## 1. Explain the difference between Relative, Absolute and Mixed Cell Referencing.

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**Relative Cell Referencing**

Cell referencing is used in Excel formulas to refer to other cells. There are three main types of cell references:

### ❖ Relative Reference

A relative reference changes automatically when a formula is copied to another cell. It is useful when performing the same calculation for multiple rows or columns.

- **Format:** A1
- **Example:**
  If formula is =A1+B1
  When dragged down → =A2+B2

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### ❖ Absolute Reference

An absolute reference remains fixed even when copied to another cell. It is used when a constant value (like tax rate or discount) must stay the same.

- **Format:** $A$1
- **Example:**
  =A1*$D$1
  When copied down → =A2*$D$1
  (Cell D1 remains constant)

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### ❖ Mixed Reference

Mixed reference locks either the row or the column, but not both.

- **Format:** $A1 (Column fixed)
- **Format:** A$1 (Row fixed)
- **Example:** =$A1*B$1

It is useful in matrix calculations or structured tables.

## 2. Write a formula to calculate the total sales of Car and Bicyle only only.

→

This formula calculates the combined sales value of only two specific products: Car and Bicycle.

✅ **Formula:**

=SUMIF(A2:A4,"Car",B2:B4) + SUMIF(A2:A4,"Bicycle",B2:B4)

📌 **Description:**

The SUMIF function adds values based on a given condition.
Here, it checks the vehicle name column and adds sales only where the product is Car or Bicycle.

D2:E10 ▾ | fx =SUMIF(A2:A4,"Car",B2:B4) + SUMIF(A2:A4,"Bicycle",B2:B4)

| | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| 1 | Vehicle | Price | | | | |
| 2 | Car | 500000 | | 510500 | | |
| 3 | Bicycle | 10500 | | | | |
| 4 | Auto-Rickshaw | 25000 | | | | |
| 5 | | | | | | |
| 6 | | | | | | |
| 7 | | | | | | |
| 8 | | | | | | |
| 9 | | | | | | |
| 10 | | | | | | |
| 11 | | | | | | |
| 12 | | | | | | |
| 13 | | | | | | |
| 14 | | | | | | |

ExcelLink: [https://drive.google.com/drive/u/0/folders/1EHyhMLiKDw_MMq-2wyRYqL0N3Zly816V](https://drive.google.com/drive/u/0/folders/1EHyhMLiKDw_MMq-2wyRYqL0N3Zly816V)

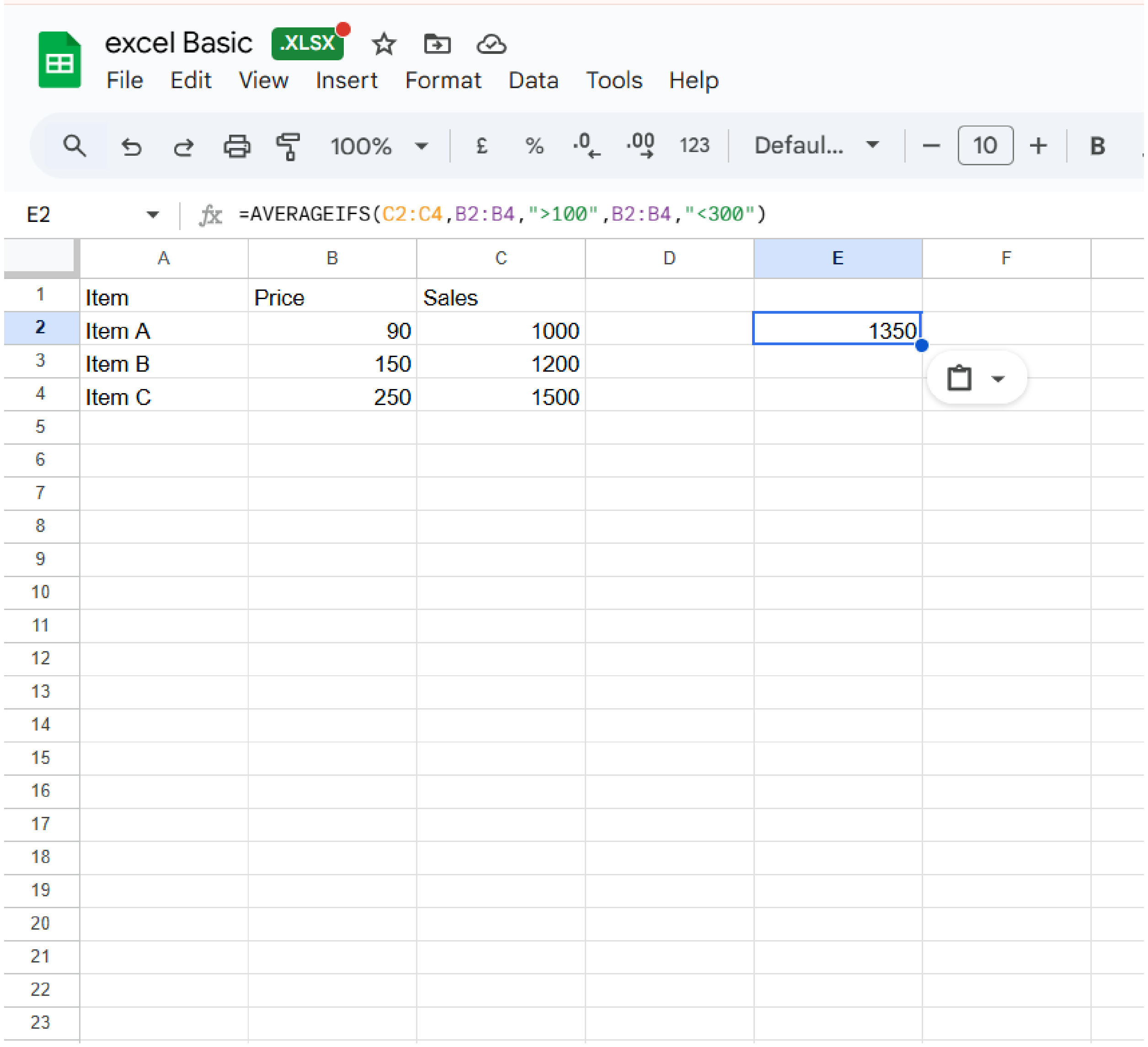
## 3) Using the data below, write a formula to calculate the average sales of items priced above 100 but less than 300:

→

This formula calculates the average sales amount for products within a specific price range.

- use this formula: =AVERAGEIFS(C2:C4,B2:B4,">100",B2:B4,"<300")
- AVERAGEIFS calculates the average based on multiple conditions.
  It includes only those items whose price is:
  Greater than 100

Less than 300
This is useful for filtered analysis.



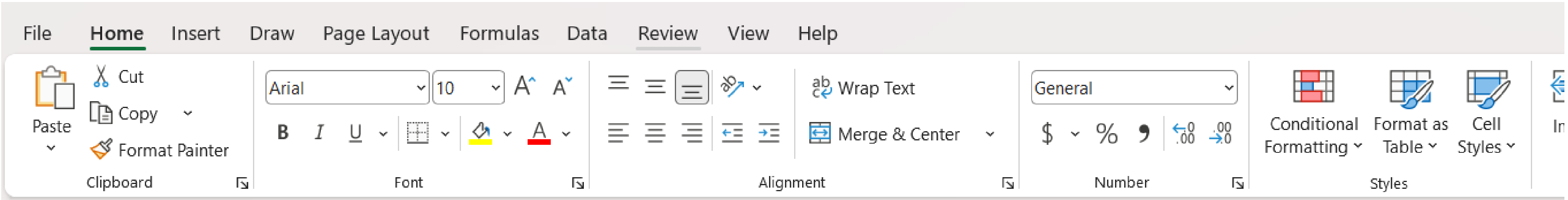

EXCEL LINK :-
https://docs.google.com/spreadsheets/d/1rhy00W9Ua2Km29aLGgWlVRqsFs251SmU/edit?gid=1723342740#gid=1723342740

**4. Count how many customer names are recorded.**
→

This formula counts the total number of customer names entered in the dataset.
Formula:- =COUNTA(A2:A100)

📌 **Description:**

The COUNTA function counts all non-empty cells.
Since customer names are text values, COUNTA is used instead of COUNT.



=COUNTA(B2:B100)

| | A | B | C | D | E | F | G | H | I |
|---|---|---|---|---|---|---|---|---|---|
| 1 | OrderID | CustomerNa | Region | Product | Quantity | UnitPrice | OrderDate | | |
| 2 | 1001 | Amit Sharma | North | Notebook | 10 | 50 | 05-01-2024 | | 50 |
| 3 | 1002 | Neha Verma | South | Pen | 20 | 10 | 06-01-2024 | | |
| 4 | 1003 | Rahul Singh | East | Marker | 15 | 25 | 07-01-2024 | | |
| 5 | 1004 | Pooja Mehta | West | Notebook | 5 | 50 | 08-01-2024 | | |
| 6 | 1005 | Ankit Jain | North | Pencil | 30 | 5 | 10-01-2024 | | |
| 7 | 1006 | Sneha Patel | South | Notebook | 12 | 50 | 11-01-2024 | | |
| 8 | 1007 | Vikas Gupta | East | Eraser | 25 | 8 | 13-01-2024 | | |
| 9 | 1008 | Meera Nair | West | Pen | 18 | 10 | 15-01-2024 | | |
| 10 | 1009 | Anjali Rao | North | Marker | 22 | 25 | 17-01-2024 | | |

5. Calculate the Total Sales for each row using a formula.

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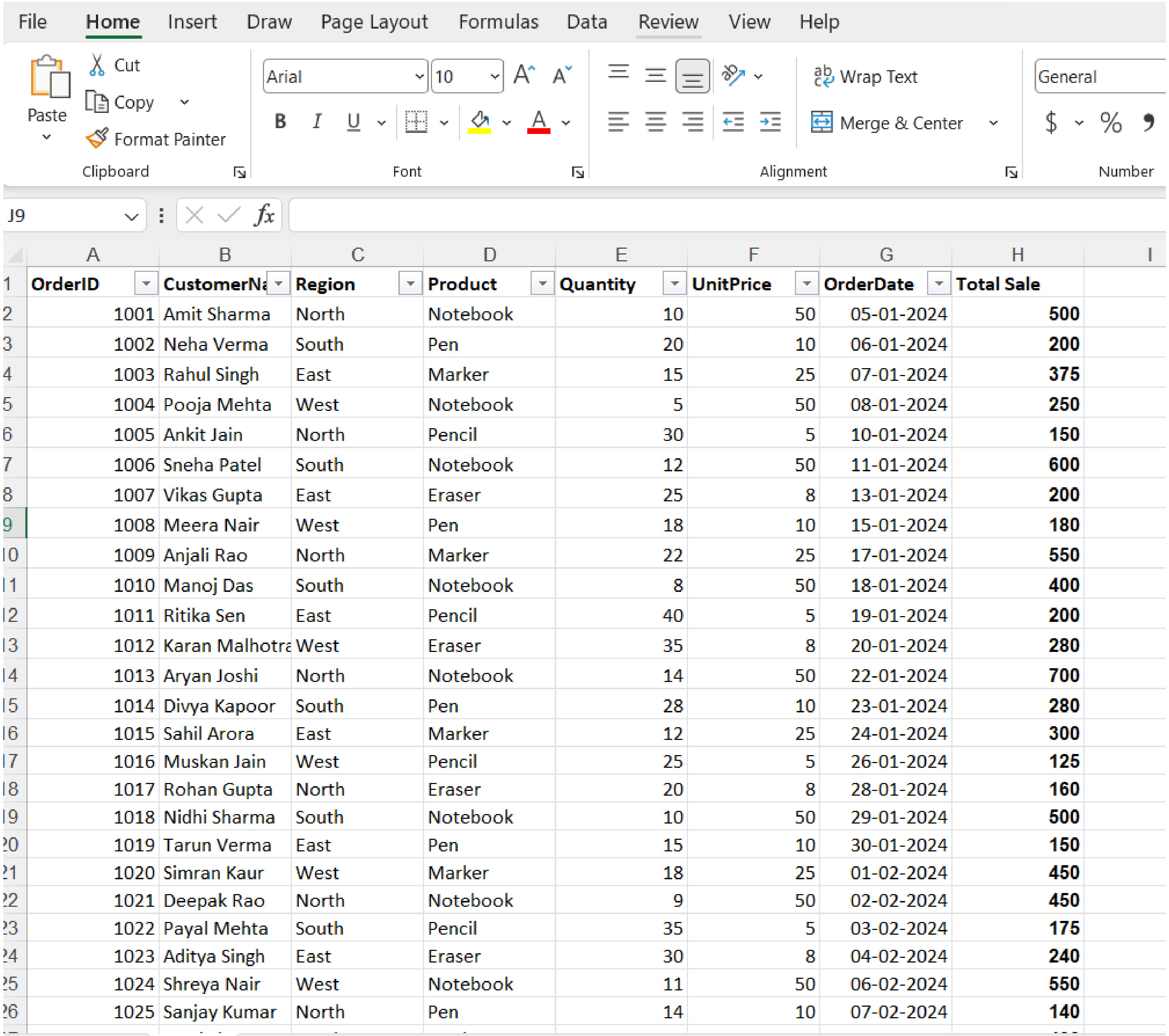
This formula calculates total sales by multiplying Quantity with Unit Price.

Formula:- =E2*F2

**Description:**

- Quantity is stored in Column E
- Unit Price is stored in Column F
- The formula multiplies both values to get total sales per transaction.
  After entering the formula in the first row, it can be dragged down to apply to all rows.




| OrderID | CustomerNa | Region | Product | Quantity | UnitPrice | OrderDate | Total Sale |
|---|---|---|---|---|---|---|---|
| 1001 | Amit Sharma | North | Notebook | 10 | 50 | 05-01-2024 | **500** |
| 1002 | Neha Verma | South | Pen | 20 | 10 | 06-01-2024 | **200** |
| 1003 | Rahul Singh | East | Marker | 15 | 25 | 07-01-2024 | **375** |
| 1004 | Pooja Mehta | West | Notebook | 5 | 50 | 08-01-2024 | **250** |
| 1005 | Ankit Jain | North | Pencil | 30 | 5 | 10-01-2024 | **150** |
| 1006 | Sneha Patel | South | Notebook | 12 | 50 | 11-01-2024 | **600** |
| 1007 | Vikas Gupta | East | Eraser | 25 | 8 | 13-01-2024 | **200** |
| 1008 | Meera Nair | West | Pen | 18 | 10 | 15-01-2024 | **180** |
| 1009 | Anjali Rao | North | Marker | 22 | 25 | 17-01-2024 | **550** |
| 1010 | Manoj Das | South | Notebook | 8 | 50 | 18-01-2024 | **400** |
| 1011 | Ritika Sen | East | Pencil | 40 | 5 | 19-01-2024 | **200** |
| 1012 | Karan Malhotra | West | Eraser | 35 | 8 | 20-01-2024 | **280** |
| 1013 | Aryan Joshi | North | Notebook | 14 | 50 | 22-01-2024 | **700** |
| 1014 | Divya Kapoor | South | Pen | 28 | 10 | 23-01-2024 | **280** |
| 1015 | Sahil Arora | East | Marker | 12 | 25 | 24-01-2024 | **300** |
| 1016 | Muskan Jain | West | Pencil | 25 | 5 | 26-01-2024 | **125** |
| 1017 | Rohan Gupta | North | Eraser | 20 | 8 | 28-01-2024 | **160** |
| 1018 | Nidhi Sharma | South | Notebook | 10 | 50 | 29-01-2024 | **500** |
| 1019 | Tarun Verma | East | Pen | 15 | 10 | 30-01-2024 | **150** |
| 1020 | Simran Kaur | West | Marker | 18 | 25 | 01-02-2024 | **450** |
| 1021 | Deepak Rao | North | Notebook | 9 | 50 | 02-02-2024 | **450** |
| 1022 | Payal Mehta | South | Pencil | 35 | 5 | 03-02-2024 | **175** |
| 1023 | Aditya Singh | East | Eraser | 30 | 8 | 04-02-2024 | **240** |
| 1024 | Shreya Nair | West | Notebook | 11 | 50 | 06-02-2024 | **550** |
| 1025 | Sanjay Kumar | North | Pen | 14 | 10 | 07-02-2024 | **140** |

6. Calculate the total sales of Notebooks in the North region only.
→This formula calculates sales only for a specific product (Notebook) in a specific region (North).

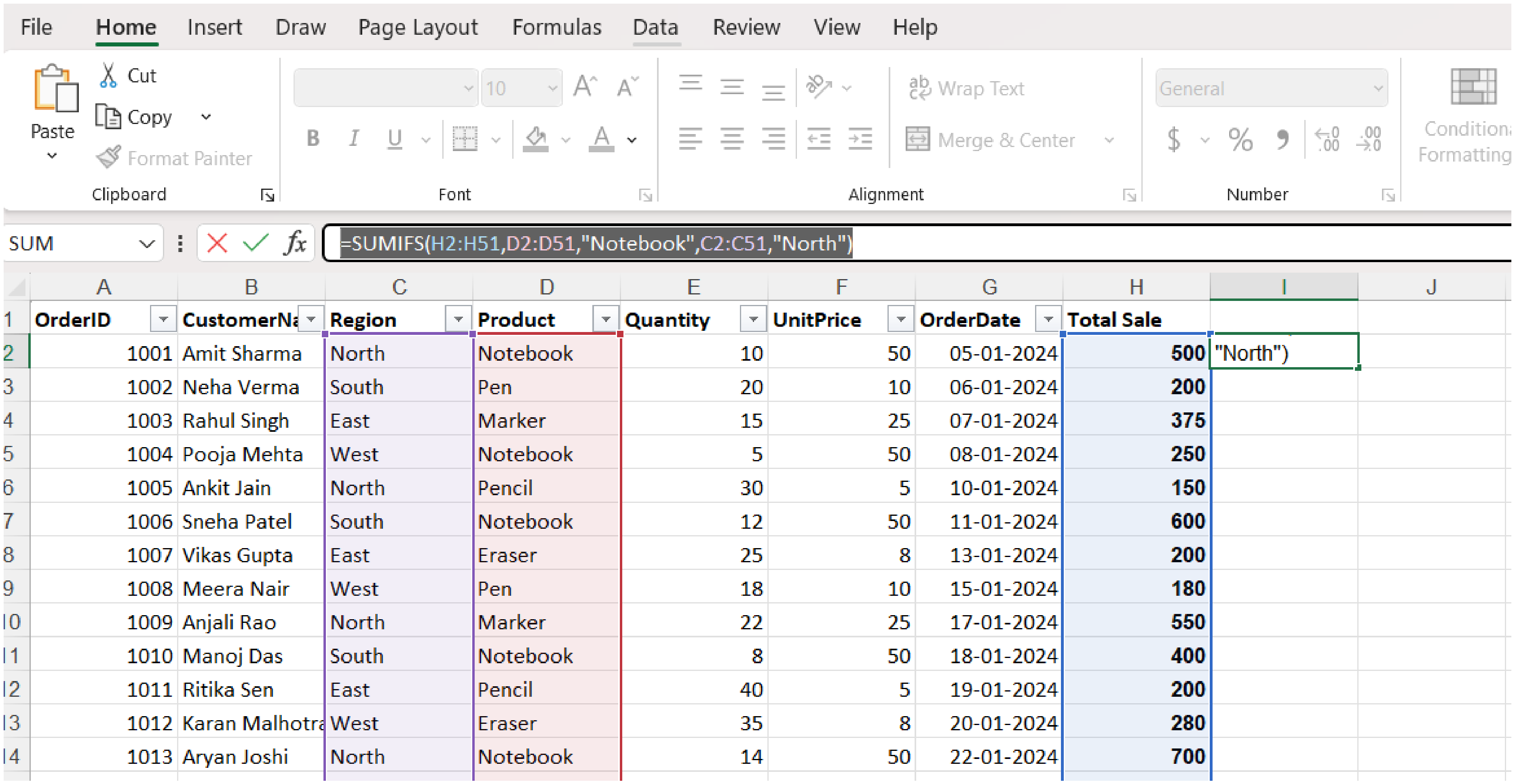
✅ **Formula:**

=SUMIFS(H2:H51,D2:D51,"Notebook",C2:C51,"North")

📌 **Description:**

SUMIFS allows multiple conditions:

- Product must be “Notebook”
- Region must be “North”
- Sales values are added from the Total Sales column




| | A | B | C | D | E | F | G | H | I | J |
|---|---|---|---|---|---|---|---|---|---|---|
| 1 | OrderID | CustomerNa | Region | Product | Quantity | UnitPrice | OrderDate | Total Sale | | |
| 2 | 1001 | Amit Sharma | North | Notebook | 10 | 50 | 05-01-2024 | 500 | "North") | |
| 3 | 1002 | Neha Verma | South | Pen | 20 | 10 | 06-01-2024 | 200 | | |
| 4 | 1003 | Rahul Singh | East | Marker | 15 | 25 | 07-01-2024 | 375 | | |
| 5 | 1004 | Pooja Mehta | West | Notebook | 5 | 50 | 08-01-2024 | 250 | | |
| 6 | 1005 | Ankit Jain | North | Pencil | 30 | 5 | 10-01-2024 | 150 | | |
| 7 | 1006 | Sneha Patel | South | Notebook | 12 | 50 | 11-01-2024 | 600 | | |
| 8 | 1007 | Vikas Gupta | East | Eraser | 25 | 8 | 13-01-2024 | 200 | | |
| 9 | 1008 | Meera Nair | West | Pen | 18 | 10 | 15-01-2024 | 180 | | |
| 10 | 1009 | Anjali Rao | North | Marker | 22 | 25 | 17-01-2024 | 550 | | |
| 11 | 1010 | Manoj Das | South | Notebook | 8 | 50 | 18-01-2024 | 400 | | |
| 12 | 1011 | Ritika Sen | East | Pencil | 40 | 5 | 19-01-2024 | 200 | | |
| 13 | 1012 | Karan Malhotra | West | Eraser | 35 | 8 | 20-01-2024 | 280 | | |
| 14 | 1013 | Aryan Joshi | North | Notebook | 14 | 50 | 22-01-2024 | 700 | | |

7. Create a column chart showing total sales by product.

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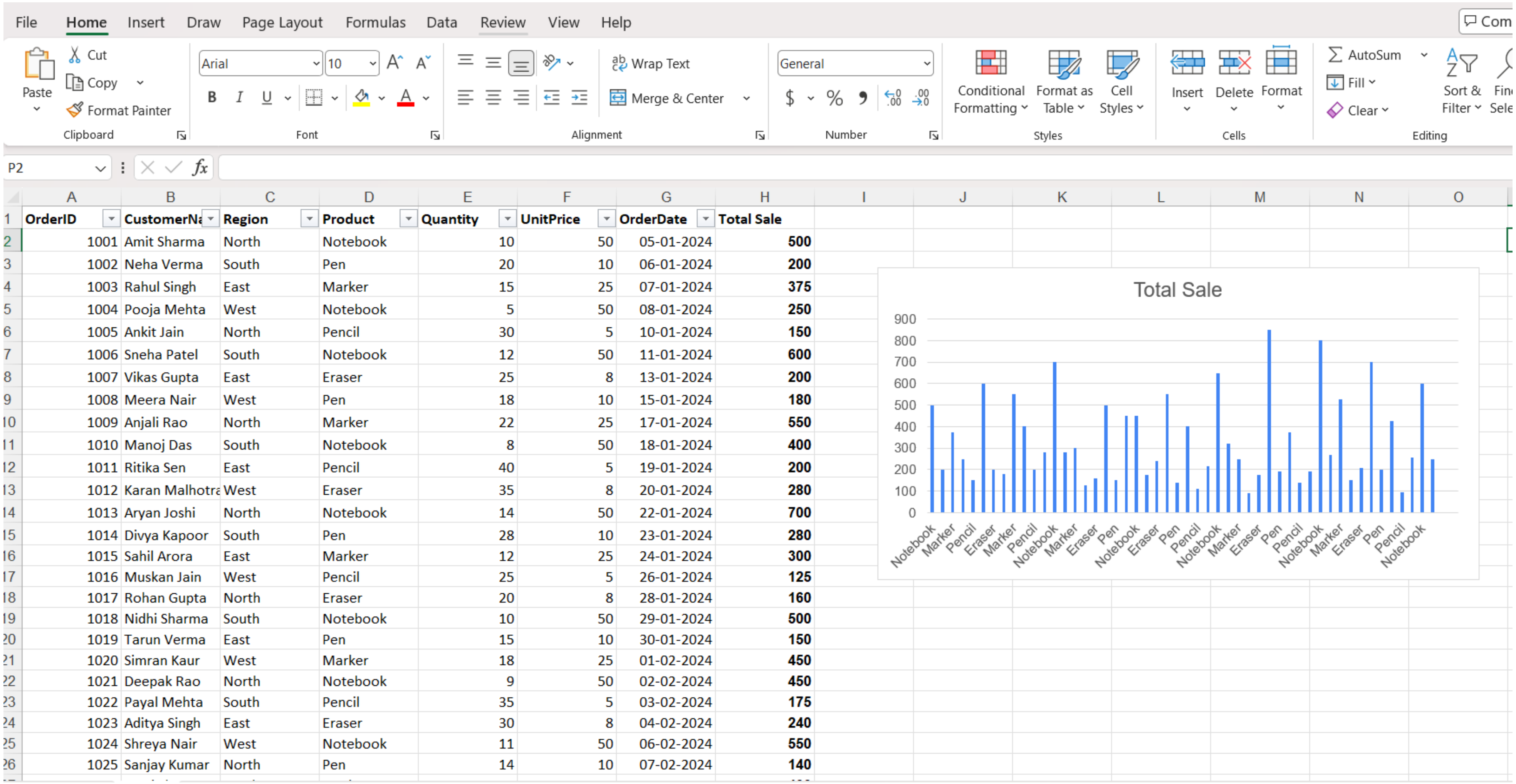
A column chart visually compares total sales of different products.

📌 **Steps:**

1. Select Product and Total Sales columns.
2. Go to **Insert Tab**.
3. Choose **Clustered Column Chart**.
4. Add title: **"Total Sales by Product"**.

## Description:

5. Column charts help in comparing performance between products clearly and effectively.




| OrderID | CustomerNa | Region | Product | Quantity | UnitPrice | OrderDate | Total Sale |
|---|---|---|---|---|---|---|---|
| 1001 | Amit Sharma | North | Notebook | 10 | 50 | 05-01-2024 | 500 |
| 1002 | Neha Verma | South | Pen | 20 | 10 | 06-01-2024 | 200 |
| 1003 | Rahul Singh | East | Marker | 15 | 25 | 07-01-2024 | 375 |
| 1004 | Pooja Mehta | West | Notebook | 5 | 50 | 08-01-2024 | 250 |
| 1005 | Ankit Jain | North | Pencil | 30 | 5 | 10-01-2024 | 150 |
| 1006 | Sneha Patel | South | Notebook | 12 | 50 | 11-01-2024 | 600 |
| 1007 | Vikas Gupta | East | Eraser | 25 | 8 | 13-01-2024 | 200 |
| 1008 | Meera Nair | West | Pen | 18 | 10 | 15-01-2024 | 180 |
| 1009 | Anjali Rao | North | Marker | 22 | 25 | 17-01-2024 | 550 |
| 1010 | Manoj Das | South | Notebook | 8 | 50 | 18-01-2024 | 400 |
| 1011 | Ritika Sen | East | Pencil | 40 | 5 | 19-01-2024 | 200 |
| 1012 | Karan Malhotra | West | Eraser | 35 | 8 | 20-01-2024 | 280 |
| 1013 | Aryan Joshi | North | Notebook | 14 | 50 | 22-01-2024 | 700 |
| 1014 | Divya Kapoor | South | Pen | 28 | 10 | 23-01-2024 | 280 |
| 1015 | Sahil Arora | East | Marker | 12 | 25 | 24-01-2024 | 300 |
| 1016 | Muskan Jain | West | Pencil | 25 | 5 | 26-01-2024 | 125 |
| 1017 | Rohan Gupta | North | Eraser | 20 | 8 | 28-01-2024 | 160 |
| 1018 | Nidhi Sharma | South | Notebook | 10 | 50 | 29-01-2024 | 500 |
| 1019 | Tarun Verma | East | Pen | 15 | 10 | 30-01-2024 | 150 |
| 1020 | Simran Kaur | West | Marker | 18 | 25 | 01-02-2024 | 450 |
| 1021 | Deepak Rao | North | Notebook | 9 | 50 | 02-02-2024 | 450 |
| 1022 | Payal Mehta | South | Pencil | 35 | 5 | 03-02-2024 | 175 |
| 1023 | Aditya Singh | East | Eraser | 30 | 8 | 04-02-2024 | 240 |
| 1024 | Shreya Nair | West | Notebook | 11 | 50 | 06-02-2024 | 550 |
| 1025 | Sanjay Kumar | North | Pen | 14 | 10 | 07-02-2024 | 140 |

8. Insert a line chart showing daily sales trend.

A line chart is used to show sales performance over time.

## Steps:

1. Select Date and Total Sales columns.
2. Go to **Insert Tab**.
3. Choose **2-D Line Chart**.
4. Add title: **"Daily Sales Trend"**.

## Description:

Line charts are ideal for identifying trends, growth patterns, and fluctuations in daily sales.

| OrderID | CustomerNa | Region | Product | Quantity | UnitPrice | OrderDate | Total Sale |
|---|---|---|---|---|---|---|---|
| 1001 | Amit Sharma | North | Notebook | 10 | 50 | 05-01-2024 | **500** |
| 1002 | Neha Verma | South | Pen | 20 | 10 | 06-01-2024 | **200** |
| 1003 | Rahul Singh | East | Marker | 15 | 25 | 07-01-2024 | **375** |
| 1004 | Pooja Mehta | West | Notebook | 5 | 50 | 08-01-2024 | **250** |
| 1005 | Ankit Jain | North | Pencil | 30 | 5 | 10-01-2024 | **150** |
| 1006 | Sneha Patel | South | Notebook | 12 | 50 | 11-01-2024 | **600** |
| 1007 | Vikas Gupta | East | Eraser | 25 | 8 | 13-01-2024 | **200** |
| 1008 | Meera Nair | West | Pen | 18 | 10 | 15-01-2024 | **180** |
| 1009 | Anjali Rao | North | Marker | 22 | 25 | 17-01-2024 | **550** |
| 1010 | Manoj Das | South | Notebook | 8 | 50 | 18-01-2024 | **400** |
| 1011 | Ritika Sen | East | Pencil | 40 | 5 | 19-01-2024 | **200** |
| 1012 | Karan Malhotra | West | Eraser | 35 | 8 | 20-01-2024 | **280** |
| 1013 | Aryan Joshi | North | Notebook | 14 | 50 | 22-01-2024 | **700** |
| 1014 | Divya Kapoor | South | Pen | 28 | 10 | 23-01-2024 | **280** |
| 1015 | Sahil Arora | East | Marker | 12 | 25 | 24-01-2024 | **300** |
| 1016 | Muskan Jain | West | Pencil | 25 | 5 | 26-01-2024 | **125** |
| 1017 | Rohan Gupta | North | Eraser | 20 | 8 | 28-01-2024 | **160** |
| 1018 | Nidhi Sharma | South | Notebook | 10 | 50 | 29-01-2024 | **500** |
| 1019 | Tarun Verma | East | Pen | 15 | 10 | 30-01-2024 | **150** |
| 1020 | Simran Kaur | West | Marker | 18 | 25 | 01-02-2024 | **450** |
| 1021 | Deepak Rao | North | Notebook | 9 | 50 | 02-02-2024 | **450** |
| 1022 | Payal Mehta | South | Pencil | 35 | 5 | 03-02-2024 | **175** |
| 1023 | Aditya Singh | East | Eraser | 30 | 8 | 04-02-2024 | **240** |
| 1024 | Shreya Nair | West | Notebook | 11 | 50 | 06-02-2024 | **550** |